

Market and Patent Research Report.

ProdDev SS25 - Group A

April 14, 2025

1 Introduction

Invasive alien plant species such as *Heracleum mantegazzianum*, *Lupinus polyphyllus*, and *Jacobaea vulgaris* pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing. Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2 Market Overview and Technological Landscape

2.1 RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum*.
- **Advantages:**
 - Non-chemical and environmentally safe.
 - Minimal soil disturbance.
 - Suitable for robotic integration. [Carbon Robotics](#)
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.



Figure 1: Source: RootWave

Video Demonstration:

- [RootWave Demonstration 1](#)
- [RootWave Demonstration 2](#)

2.2 Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO₂ lasers for precision weed eradication.
- **Target Species:** More suitable for row crops. Not designed specifically for tall or toxic plants like hogweed.



Figure 2: Source: RootWave

- **Video Demonstration:** [LaserWeeder Overview](#)

2.3 Odd. Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Target Species:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.



Figure 3: Source: RootWave

2.4 Drone and Aerial Detection Platforms (DK, DE, UK)

Developer: University of Connecticut (BirdHabitatBot)

- **Technology:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Advantages:** Ideal for pre-mapping and targeting robotic ground intervention.



Figure 4: Source: RootWave

2.5 FarmWise – Titan FT-35 (USA)

- **Manufacturer:** [FarmWise](#)
- **Technology:** A self-driving robotic cultivator with vision and mechanical blade system for precision weeding. Optional fertilizer or pesticide application.
- **Target Species:** Broadleaf weeds in vegetable crops.
- **Use Cases:** Commercial vegetable farms.

2.6 Naïo Technologies – Dino (France)

- **Manufacturer:** [Naïo Technologies](#)
- **Technology:** GPS navigation, laser scanners, vision-based selective mechanical weeding.
- **Target Species:** Weeds between and near crop rows.
- **Use Cases:** Organic farms requiring non-chemical weeding.

2.7 ecoRobotix – AVO (Switzerland)

- **Manufacturer:** [ecoRobotix](#)
- **Technology:** Solar-powered robot with precision herbicide spray using machine learning.
- **Target Species:** Isolated weeds among broadleaf crops.
- **Use Cases:** Sustainable farming.

2.8 FarmDroid – FD20 (Denmark)

- **Manufacturer:** [FarmDroid](#)
- **Technology:** GPS-based seeding and blind mechanical weeding.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** Organic grain and vegetable farms.

2.9 Small Robot Company – “Tom & Dick” (UK)

- **Manufacturer:** [Small Robot Company](#)
- **Technology:** Tom scouts and maps, Dick applies electric shock to kill weeds.
- **Target Species:** Broadleaf weeds in high-value crops.
- **Use Cases:** Precision agriculture.

2.10 AutoWeed – Lantana Targeting Robot (Australia)

- **Developer:** [James Cook University](#)
- **Technology:** AI and cameras detect and spray herbicide on *Lantana camara*.
- **Target Species:** *Lantana camara*.
- **Use Cases:** Biodiversity protection.

2.11 BirdHabitatBot – Forest Invasives Robot (USA)

- **Developer:** [University of Connecticut](#)
- **Technology:** Drone-GPS mapping + ground robot to remove invasive shrubs.
- **Target Species:** *Japanese barberry*, *multiflora rose*.
- **Use Cases:** Forest conservation.

2.12 Bosch Deepfield – BoniRob (Germany)

- **Developer:** [Bosch Research](#)
- **Technology:** Vision-guided pneumatic “stamping” system.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** High-speed non-chemical cultivation.

2.13 Hydrilla Hunter – Robotic Boat (USA)

- **Developer:** [Northeastern University & CAES](#)
- **Technology:** Hyperspectral aquatic detection and robotic navigation.
- **Target Species:** *Hydrilla*, *Eurasian watermilfoil*.
- **Use Cases:** Freshwater habitat protection.

3 Patent Research Overview

3.1 EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in *Heracleum mantegazzianum*.
- **Significance:** Basis for robotic puncturing adaptation.
- **Link:** [EP3804483A1](#)

3.2 WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon Robotics
- **Description:** Robotic platform guiding laser source for weed removal.
- **Limitation:** Not optimized for toxic plants or non-row fields.
- **Link:** [WO2021055485A1](#)

3.3 US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using sensors and mechanical tools.
- **Potential Use:** Adaptable for invasive species.
- **Link:** [US20170238460A1](#)

3.4 EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH
- **Description:** Electrode-based high-voltage weed inactivation.
- **Relevance:** Integrable into autonomous platforms.
- **Link:** [EP3744173A1](#)

3.5 DE102015209879A1 – Device for Damaging Weeds

- **Assignee:** Bosch GmbH
- **Description:** Tool with classification and positioning unit for targeted weed damage.
- **Significance:** Precise, rapid chemical-free weed control.
- **Link:** [DE102015209879A1](#)

3.6 DE4039797A1 – Weed Control Device

- **Assignee:** Robert Bosch GmbH
- **Description:** Continuous weed destruction system interrupted only when crop detected.
- **Significance:** Efficient for automated systems.
- **Link:** [DE4039797A1](#)

Traide Fair Australia

During a recent trade fair visit (14.04.2025) to XAG Sydney Australia, supported by further online research, the **XAG P150 agricultural drone** was identified as a promising solution for the removal of invasive plant species. The drone features a payload capacity of up to 70 kg and uses the **RevoSpray 4 system** to apply herbicides either across wide areas or with pinpoint accuracy. The spraying system delivers fine droplets via adjustable centrifugal nozzles, enabling effective coverage on both sides of plant leaves.

The process is managed using **PIX4Dfields**, a mapping and control software. First, a drone equipped with a high-resolution camera captures detailed images of the area. In the software, the user manually marks invasive species, protected plants, and obstacles. These inputs train the software to recognize similar vegetation patterns using machine learning. The generated prescription map is then uploaded to the spraying drone, which treats only the marked zones with high precision.

Compared to ground-based mowers like the **ROBO MOWER T-90** by NetEvoTech, spraying drones offer several advantages. Mechanical mowing can lead to seed dispersion and is limited in difficult or uneven terrain. Experts advised against the use of mowers in this context and recommended drone application to avoid further spread and ensure better reach. The system is also compatible with drone charging stations from **Enovance**, which can quickly recharge batteries for continuous operation. The total cost for a two-drone system including software is approximately **AUD 50,000**.

Sources:

XAG P150 Product Page: <https://www.xa.com/en/p150>

PIX4Dfields Software Overview: <https://www.pix4d.com/product/pix4dfields>

RevoSpray 4 Spraying System: <https://www.xa.com/en/revospray>

ROBO MOWER T-90 Specs (NetEvoTech): <https://www.netevotech.com/robo-mower-t90>

Enovance Drone Charging: <https://enovance.com.au/>

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XAG® P150 Agricultural Drone

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- Delivers precise spraying and spreading, ensuring crops get exactly what they need.
- Reduce overspray, save on chemicals, and maximize crop health.
- Works when you need it — no waiting for pilots or weather windows.



XAG® SRC4 Remote Controller

Elevate your agricultural drone operations with 6-inch ultra-bright screen, 2 km range, 5-hour battery life, and real-time connectivity for precision agriculture.



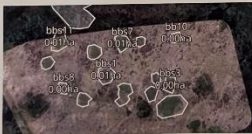
XAG® DL1 Data Link

Designed to extend your drone's communication range, eliminate signal dead zones, and ensure uninterrupted operations in even the most challenging environments.



XAG® XRTK6 Pro Mobile Station

Achieve centimetre-level accuracy with the XAG® XRTK6 Pro Mobile Station. Portable. RTK, 15 km Wi-Fi range, and offline satellite correction for precision farming.



Spot Spraying

Multi-Field Application — Ideal for small fields or targeted area applications.



Swarm Control

Swarm Operation allows for the autonomous control of two drones simultaneously.



XAG® Care P150

Fly with Peace of Mind

Comprehensive coverage for your XAG® P150, so unexpected repair costs don't slow down your farming success.

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NetEvoTech

ROBO MOWER T-PREMIUM

Stronger Build with Eco Power Option



Battery Version Available



Choose the battery-powered option of T-PREMIUM for lower noise levels and eco-friendly, emission-free performance.



Reinforced Tracks



Upgraded Dual Blades



Powerful Lift System



Debris Protection

Optional Accessories



Snow Plough



Rotary Brush



Industrial Remote Control



Remote Control with Display and Camera



Tow Hitch



Upgraded Enclosed Design

The advanced enclosed design offers superior protection, enhancing the safety and longevity of both the machine and the operator.

- **Advanced brushless electric motor:** Efficiently powers the tracks, delivering smooth, reliable movement.
- **Durable hardened steel blades:** Built to resist wear, guaranteeing long-lasting sharpness.
- **Long-range remote control:** Manage operations effortlessly from up to 200 meters away.
- **Reinforced rubber tracks:** Engineered for durability on any terrain, giving unmatched stability.
- **Robust, impact-resistant design:** Crafted to minimize damage, ensuring top-tier resilience in all conditions.
- **Self-charging hybrid drive:** Continuously powers up while the mower works, ensuring uninterrupted performance.
- **High-powered gasoline engine:** Fuels the cutting blades with strength for superior mowing efficiency.



Take your mowing to the next level with the optional automation module, enabling precise, automated navigation along pre-planned routes.



The sharp blades and powerful engine of the T-Series ensure clean, easy cuts through small shrubs.



Reinforced heavy-duty tracks with a low centre of gravity ensure stability on slopes up to 65°.



The T-Series' super-low profile and 360° rotation easily navigate under solar panels and other low obstacles.



Maneuverability allows smooth cutting under trees, without damaging fruit trees or branches.



The T-Series' powerful climbing ability ensures efficient maintenance of steep embankments.

	 T-90	 T-PREMIUM
 Dimensions:	140cm x 130cm x 63cm	140cm x 140cm x 70cm
 Weight:	315kg	360kg
 Mowing width:	90cm	90cm
 Efficiency:	3000-4500m ² /h	3000-4500m ² /h
 Engine type:	Four stroke engine	Four stroke engine
 Speed of movement:	0-5km/h	0-5km/h
 Maximum working slope:	65°	65°
 Blade lifting range:	1-20cm	1-20cm
 Maximum cutting height:	2m	2m
 Remote control range:	200m	200m

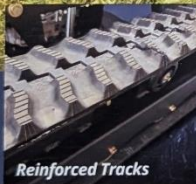


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NetEvoTech

ROBO MOWER T-90

Unleash the Power, Conquer Any Terrain



Reinforced Tracks



Upgraded Blades



Powerful Lift System



Debris Protection

Optional Accessories



Cargo Platform



Atomizer 200L



Horizontal and
Vertical Sprinkler 200L



Snow Plough



Rotary Brush



Industrial
Remote Control



Remote Control with
Display and Camera



Tow Hitch

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Overall solution for drone charging

Model: O-Power 21 (Battery Pack)



Model: IBC6000W (AC-DC Charger)

Model: IBC10000W (DC-DC Charger)



Efficient & Time-saving

Battery pack takes only 5 hours to fully charge with 220V mains power, and only 2 hours with the EV DC fast charger. Only takes about 8 minutes to charge the drone battery.



Worry-free & Reliable

300V, 6000 cycle warranty, worry-free.
Five-year warranty, free repair for non-man-made faults.
Accidental spontaneous combustion insurance up to 1 million dollars.



Silent & Zero Pollution

No noise pollution, no maintenance required.



Economical & Affordable

80% reduction in charging costs compared to gasoline generators.

O-Power 21 (Battery Pack)

Model	O-Power 21 (Battery Pack)		
Cell technology	LiFePO ₄	Nominal voltage	76.8V
Dimension (D*W*H)	820*580*295mm	Recommended crate	0.5C
Package size (D*W*H)	1300*730*540mm	USB interface voltage	5V
Weight(Net weight)	174.5kg	USB interface power	65W
Nominal capacity	280Ah	Display	4.3 inches
Nominal energy	21.5kWh	Heat dissipation method	Air cooling
Battery group mode	1P24S	Ingress protection	IP20
Rated charge/discharge current	140A	Working altitude	≤3000m
Max.Charge/Discharge current	200A	Operating temperature	0-50°C
Operating voltage	67.2V - 84.24V	Cycle life	≥3000 Cycles

IBC6000W (AC-DC Charger)

Model	IBC6000W (AC-DC Charger)		
Dimension (D*W*H)	450*223.4*66.2mm	Power	6kW
Weight (Net weight)	8.5 kg	Input method	L+N+PE
Rated input voltage	220Vac	Heat dissipation method	Air cooling
Input voltage range	90Vac-264Vac (90 - 185Vac derating 2000W output)	Ingress protection rating	IP20
Output voltage range	60Vdc - 87.6Vdc	Working altitude	≤3000m
Output current	0 - 70A	Operating temperature	0-50°C

IBC10000W (DC-DC Charger)

Model	IBC10000W (DC-DC Charger)		
Dimension (D*W*H)	550*230*94.5mm	Output power	10kW
Weight (Net weight)	14.5kg	Heat dissipation method	Air cooling
Input voltage range	65Vdc-100Vdc	Ingress protection	IP20
Output voltage range	0.5Vdc-58.8Vdc	Working altitude	≤3000m
Output current	0 - 180A (Default 150A)	Operating temperature	0-50°C



XAG® P150

RevoCast P4

XAG
AUSTRALIA

Effortless precision,
unmatched productivity



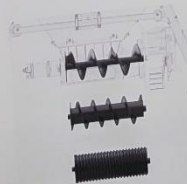
115 L
Granule
Container

280 Kg/min
Max.
Spread Rate

3-8 m'
Spread
Swath

70 Kg
Payload

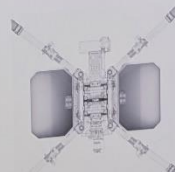
2,167 Kg/hr²
Spreading Efficiency



Versatile
Screw Feeders



Vertical
Waving Spreader



Dual-Side
Large Filling Ports



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XAG
Re

Pione
for su

18 m/s
Max. Flight

70 L Smart Liq

Spraying Efficiency. Th
in open fields. Actual p



XAG® P150

RevoSpray P4

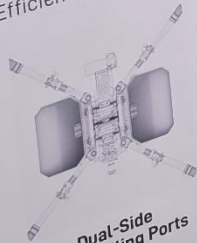


Pioneering intelligent rotary atomisation
for superior crop protection in any environment



3-8 m¹
Spread
Swath

7 kg/hr²
Efficiency



**Dual-Side
Large Filling Ports**

Factors, temperature, and user handling.
conditions.
ing, and 18 m/s speed in open fields.

18 m/s
Max. Flight Speed

30 L/min
Max. Flow Rate

5-10 m
Spraying Width

26 ha/h¹
Spraying Efficiency

60-500 µm
Droplet Size Range



70 L Smart Liquid Tank



Advanced Centrifugal Nozzles



Flexible Impeller Pumps

Spraying Efficiency: The spraying efficiency is measured at a 20 L/Ha application volume, 10 m spacing, and 18 m/s speed in open fields. Actual performance may vary due to real-life conditions.



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XAG® P150

up to **26** ha/h

OVERALL SPRAYING
EFFICIENCY¹

Older Model

up to **19** ha/h

OVERALL SPRAYING
EFFICIENCY²

≈ more than a **36% increase in productivity**



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valuation on your current
models trade in!

COMPARISON WITH OLDER AGRICULTURAL DRONES

	Tank Size	Max Operational Speed	Max Flow Rate
P150	70L	18 m/s	30 L/min
P100 Pro	50L	13.8 m/s	22 L/min
P100	40L	13.8 m/s	12 L/min
V40	16L	10 m/s	10 L/min
P30	16L	8 m/s	5.6 L/min
Other	40L	10 m/s	12 L/min

Spraying Efficiency: The spraying efficiency is measured at a 20 L/Ha application volume, 10 m spacing, and 18 m/s speed in open fields. Actual performance may vary due to real-life conditions.
19 ha/h is for normal, large-volume spraying (such as wheat pest control). P100 Pro operating parameters: 30 L/Ha, 9 m flight route spacing, 13.8 m/s flight speed.

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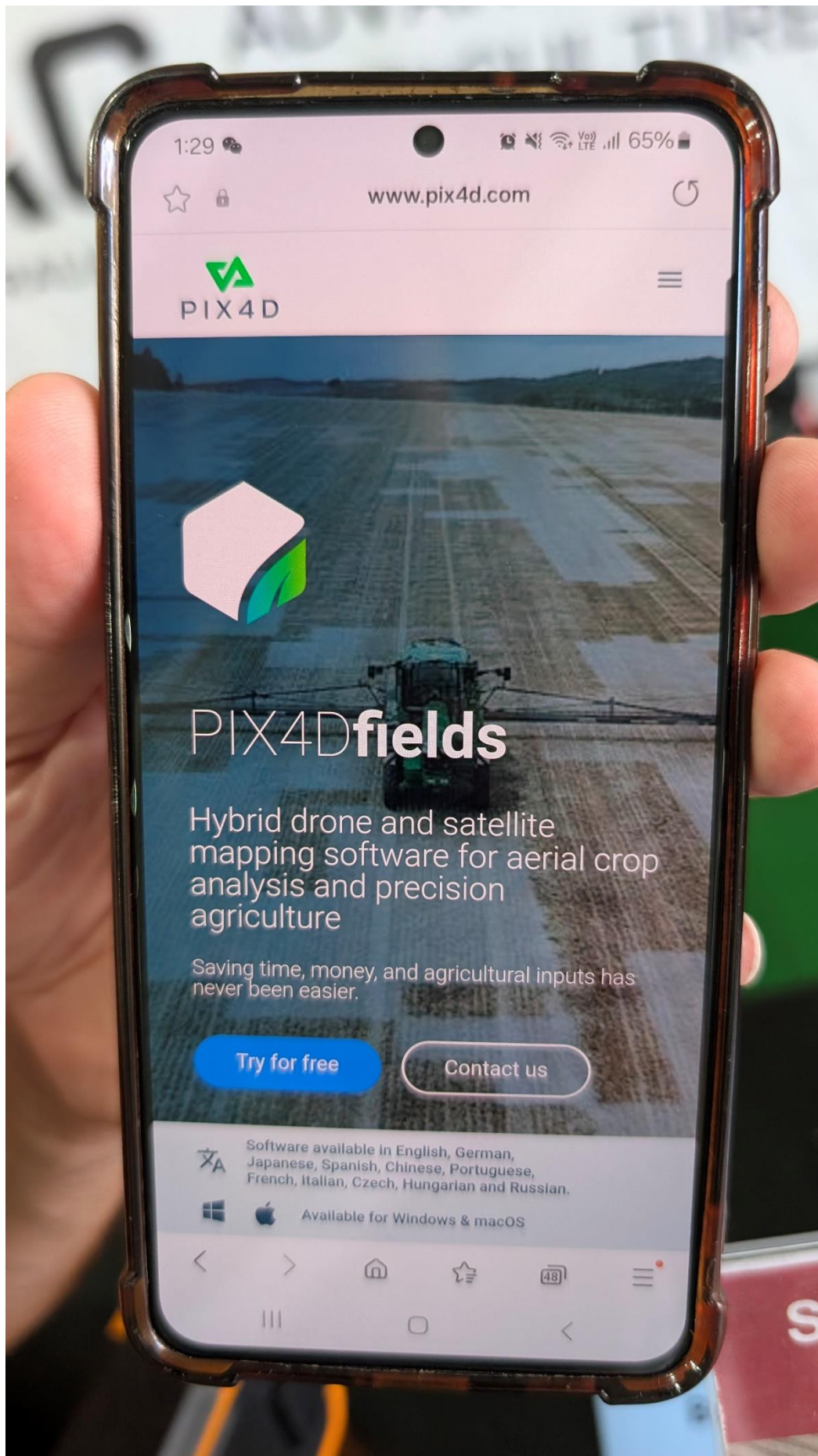
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French, Italian, Czech, Hungarian and Russian.



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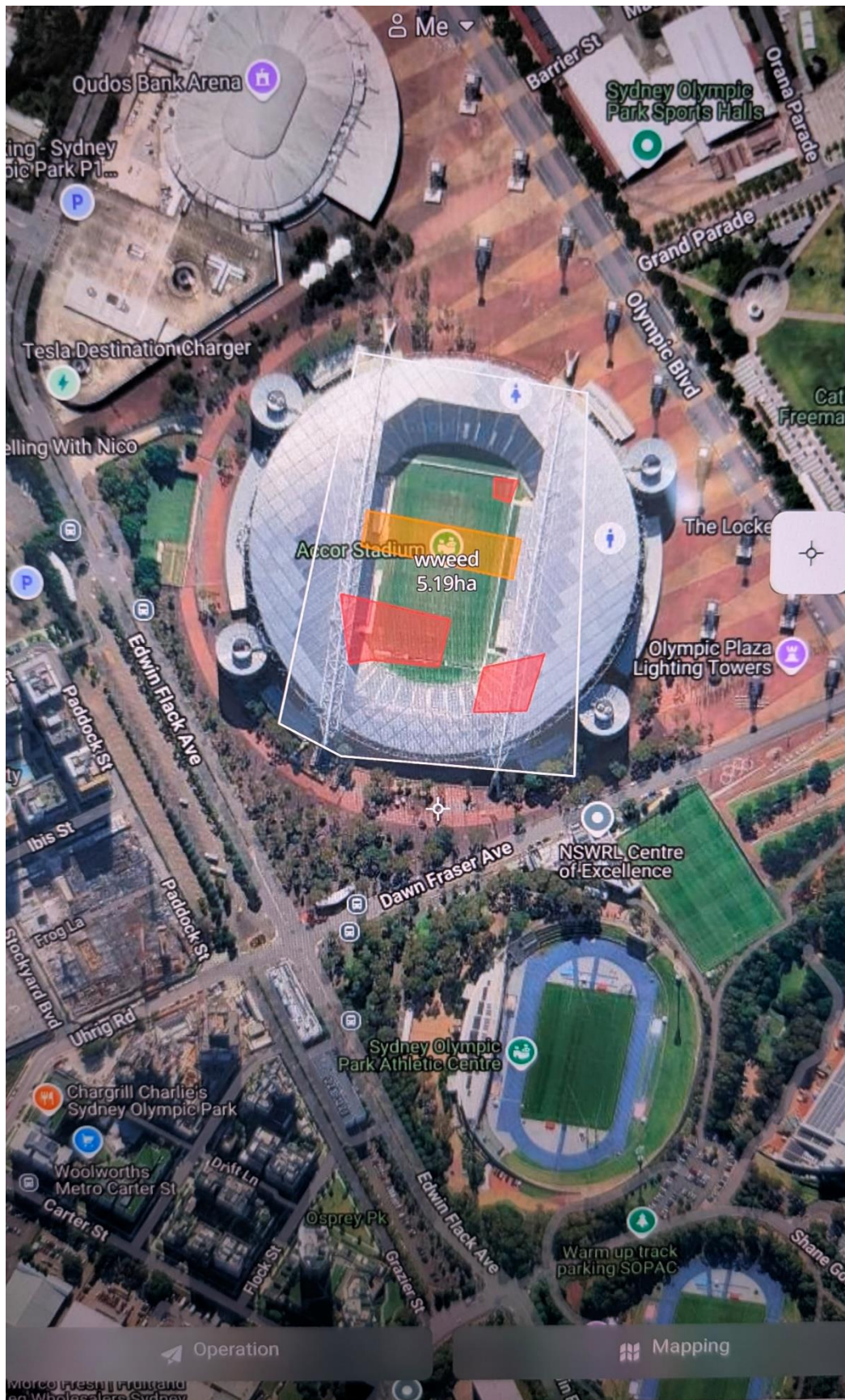


Software available in English, German,
Japanese, Spanish, Chinese, Portuguese,
French, Italian, Czech, Hungarian and Russian.



Available for Windows & macOS





This is how the App looks... Red is no go area and orange is spray area